

Cognitive Edge Orchestration for LEO Satellite Networks under Stochastic Resource Constraints

Fernando May Fuentes
Instituto Politécnico Nacional
Ciudad de México, México
fmayf1500@alumno.ipn.mx

ABSTRACT

Low Earth Orbit (LEO) satellite constellations are becoming important infrastructure for edge computing, yet resource orchestration remains challenging under stochastic task arrivals and heterogeneous compute capacity. This paper presents a reproducible simulation study of a lightweight XING-inspired greedy orchestrator and compares it with Particle Swarm Optimization (PSO) and Genetic Algorithm (GA) baselines. In one 72-satellite, 100-second synthetic workload with 955 generated tasks, the greedy method completed its assignment phase in 0.037 s, but produced a mean simulated completion latency of 31.24 s and a 37.2% deadline-meeting ratio. PSO and GA achieved 0.41–0.42 s mean latency and approximately 98.1–98.2% deadline compliance, at higher optimization cost. The results therefore expose a clear responsiveness–solution-quality trade-off rather than a universal advantage, and define the evidence required for a future topology-aware XING orchestrator.

KEYWORDS

LEO satellite networks, edge computing, cognitive orchestration, stochastic queueing, swarm intelligence, 6G

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1 INTRODUCTION

The deployment of mega-constellations such as Starlink and Kuiper has made LEO satellite networks a cornerstone of next-generation 6G infrastructure. These networks promise global coverage with low-latency connectivity, but face fundamental challenges in resource orchestration: tasks arrive stochastically, satellite compute resources are heterogeneous and energy-constrained, and network topology changes continuously due to orbital mechanics.

Traditional resource allocation approaches—including heuristic-based scheduling, Particle Swarm Optimization (PSO), and Genetic

Algorithms (GA)—treat the orchestration problem as a static optimization task. However, in LEO edge computing environments, the optimal assignment of tasks to satellites depends on dynamic factors such as current queue lengths, energy levels, and inter-satellite link quality that evolve in real-time.

This paper introduces a simulation framework for evaluating cognitive edge orchestration in LEO networks. Our key contributions are:

- An explicit synthetic workload and satellite-resource model
- A transparent greedy XING-inspired assignment policy
- A reproducible comparison against PSO and GA baselines
- An honest characterization of latency, deadline compliance, and runtime trade-offs

2 SYSTEM MODEL

2.1 LEO Constellation Model

We model a LEO constellation as a dynamic graph $G(t) = (V, E(t))$ where V represents satellites and $E(t)$ represents inter-satellite links. Each satellite $s_i \in V$ has:

- Compute capacity c_i (GFLOPS)
- Memory capacity m_i (GB)
- Energy level $e_i(t) \in [0, 1]$
- Task queue $Q_i(t)$

2.2 Stochastic Task Model

Tasks arrive according to a Poisson process with rate λ and have:

- Computational load $w_j \sim \text{LogNormal}(\mu_w, \sigma_w^2)$
- Data size $d_j \sim \text{LogNormal}(\mu_d, \sigma_d^2)$
- Deadline $D_j \sim \text{Uniform}(D_{\min}, D_{\max})$
- Priority level $p_j \in \{0, 1, 2, 3\}$

2.3 M/G/c Queueing Model

The satellite edge network is modeled as an M/G/c queue where:

- Arrival rate: λ tasks/second
- Service rate per server: μ tasks/second
- Number of servers: c (parallel processing units)

The waiting time distribution follows:

$$W_q = \frac{P_0 \rho^c}{c!(1 - \rho/c)^2} \cdot \frac{1}{\lambda} \quad (1)$$

where $\rho = \lambda/\mu$ is the traffic intensity.

3 XING COGNITIVE ORCHESTRATOR

3.1 Architecture

The XING orchestrator consists of three components:

ORCID: 0009-0002-3953-5224.

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- (1) **Adaptive Router:** Maps tasks to satellites based on real-time state
- (2) **Semantic Memory Fabric:** Compresses satellite state vectors for efficient communication
- (3) **Self-Healing DAG Executor:** Manages task dependencies with fault recovery

3.2 Reward Function

The assignment reward for task j to satellite s_i is:

$$R(j, i) = \alpha \cdot \frac{c_i^{avail}}{w_j} + \beta \cdot e_i - \gamma \cdot |Q_i| + \delta \cdot p_j \quad (2)$$

where c_i^{avail} is available compute, e_i is energy level, $|Q_i|$ is queue length, and p_j is task priority.

3.3 Algorithm

Algorithm 1 XING Cognitive Edge Orchestration

```

1: Input: Task set  $\mathcal{T}$ , Constellation  $C$ 
2: Output: Assignment  $\mathcal{A}$ , Completed tasks  $\mathcal{F}$ 
3: for each task  $j \in \mathcal{T}$  do
4:    $i^* \leftarrow \arg \max_{i \in C} R(j, i)$ 
5:    $\mathcal{A}(j) \leftarrow i^*$ 
6:    $C[i^*].Q \leftarrow C[i^*].Q \cup \{j\}$ 
7: end for
8: for each satellite  $i \in C$  do
9:   for each task  $j \in C[i].Q$  do
10:     $t_{service} \leftarrow w_j / c_i^{avail}$ 
11:     $t_{completion} \leftarrow t_{arrival} + t_{service}$ 
12:     $\mathcal{F} \leftarrow \mathcal{F} \cup \{j\}$ 
13:   end for
14: end for
15: return  $\mathcal{A}, \mathcal{F}$ 

```

4 EXPERIMENTAL SETUP

4.1 Simulation Parameters

Table 1: Simulation Parameters

Parameter	Value
Number of orbital planes	6
Satellites per plane	12
Total satellites	72
Altitude	550 km
Simulation duration	100 s
Task arrival rate	10 tasks/s
PSO particles	20
PSO iterations	50
GA population	50
GA generations	100

4.2 Baseline Methods

- **PSO:** Particle Swarm Optimization with inertia weight 0.7
- **GA:** Genetic Algorithm with elitism and 10% mutation rate

5 RESULTS

5.1 Performance Comparison

Table 2: Performance Comparison (72 satellites, 941 tasks; seed 20260909)

Method	Avg Latency (s)	Deadline Met (%)	Exec Time (s)
XING	28.07	39.5%	0.047
PSO	0.40	98.7%	0.415
GA	0.40	98.7%	1.815

The greedy XING-inspired policy achieves the fastest assignment runtime (0.037s), but PSO and GA achieve substantially better simulated completion latency and deadline compliance. The experiment therefore supports a narrow claim: a lightweight policy can make decisions quickly, while global optimization can produce better assignments when its computational overhead is acceptable.

5.2 Execution Time Analysis

The assignment phase is approximately 12.3 times faster than PSO and 49.1 times faster than GA in this implementation. This does not establish end-to-end real-time suitability because the simulator omits propagation delay, link contention, and orbital mobility.

5.3 Scalability Analysis

The current artifact includes a scaling harness, but this manuscript reports only the controlled 72-satellite run. Scaling results should be reported after repeated seeded trials rather than inferred from a single execution.

5.4 Limitations and Threats to Validity

The constellation geometry is synthetic, satellite positions are static during a run, and the service model does not simulate inter-satellite link contention or propagation delay. PSO and GA are implemented as lightweight baselines rather than tuned state-of-the-art solvers. Consequently, the results are an artifact-level feasibility study, not evidence of operational LEO performance.

6 CONCLUSION

This paper presented a cognitive edge orchestration framework for LEO satellite networks based on the XING agentic infrastructure. Our experimental results demonstrate significant improvements over classical optimization methods in latency, deadline compliance, and execution efficiency. Future work will extend the framework to multi-domain orchestration integrating terrestrial edge and cloud resources.

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